

CFD-HW9

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Outline

In this assignment, a simplified jet engine compressor will be modeled in Fluent. Specifically, the device will be simulated steadily in Fluent's Turbo machine GUI. The device will compress air via rotor blades moving at a constant angular speed (1000 rpm, 1500 rpm, 2250 rpm) into a fixed set of stator blades. Using rotational symmetry, only a partial section of the turbine will be analyzed in Fluent.

Workflow

To start, the turbo machine has to be set up. This involves declaring the rotor & stator configuration as depicted below:

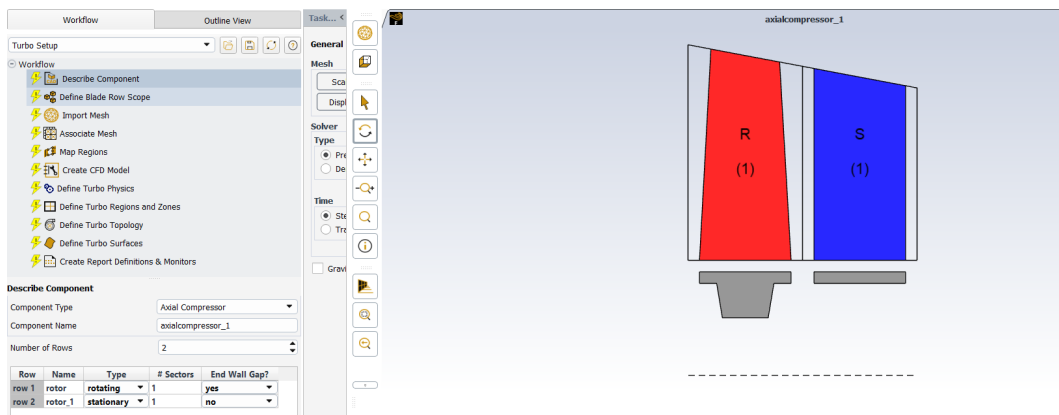


Figure 1: Turbo Machine Configuration

Next item to cover is the Mesh. The provided mesh file for the compressor section was imported into the Turbo Workflow, and the was divided according to the rotor & stator components (via Associated Mesh & Map Regions) using the default settings as depicted below:

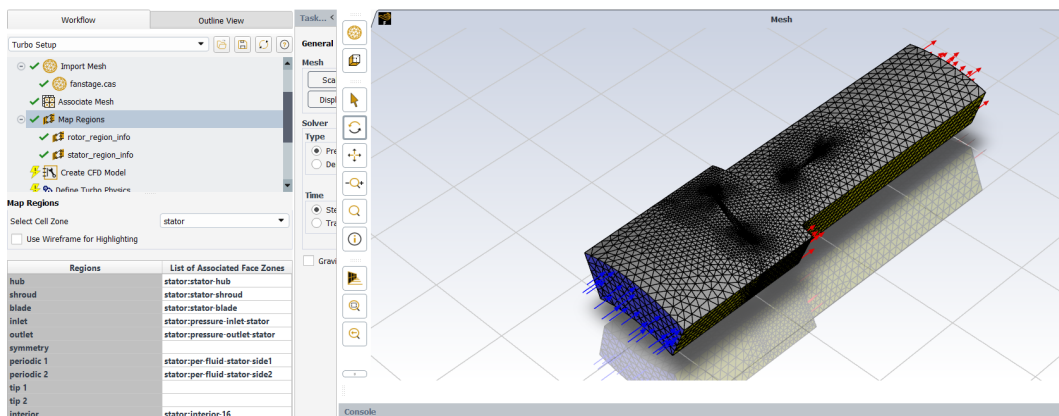


Figure 2: Mesh Decomposition

At this point, the rotor's section outlet had to be (Turbo) merged with the stator's section inlet to differentiate the rotating and static regions; this was achieved with the Turbo create feature.

Again, using the default setting, the CFD model was created, and the angular speed of the rotor blade was inputted. To start, with an operating pressure of 0 Pa, the angular speed was set to...

$$\omega = 1000 \text{ rpm}$$

The Turbo Physics were thus defined, and we could move on to the region definitions in which the default settings (pressure & temperature) were used (i.e. $p = 101325 \text{ Pa}$, $T = 300 \text{ K}$). The default topology settings and iso-surfaces were then defined. It was at this point ready for initialization and to be simulated.

The solver was set to coupled k- ϵ (standard) with 250 iterations, and the residual plot is shown below:

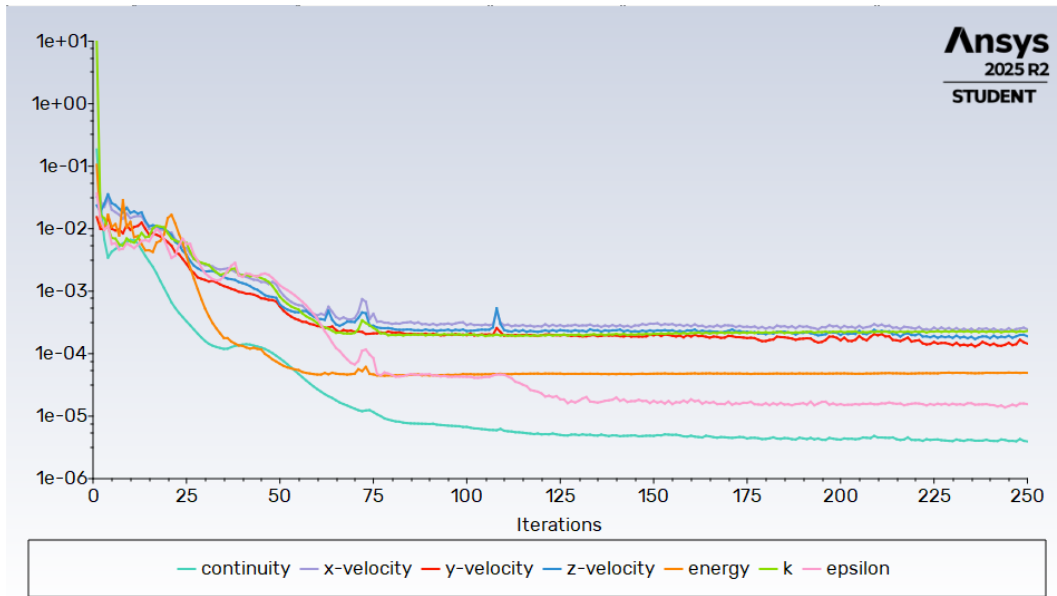


Figure 3: Residual Plot ($\omega = 1000 \text{ rpm}$)

The following contours are displayed on the (curved) mid-surface that passes through the section. The first contour to be addressed will be pressure:

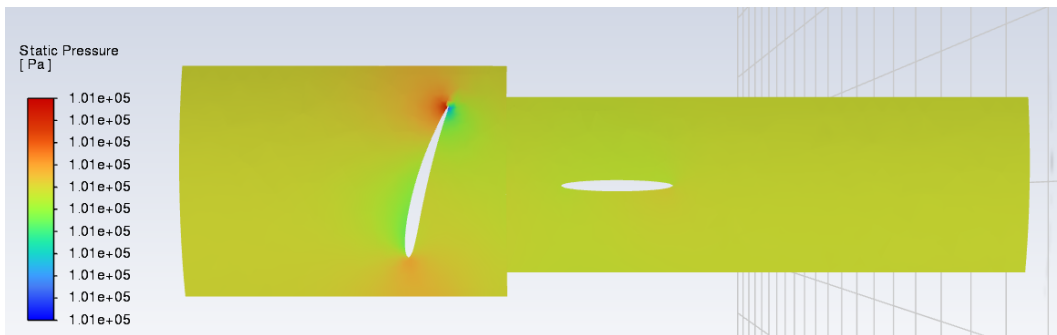


Figure 4: Pressure Contour ($\omega = 1000 \text{ rpm}$)

The (average) pressure at the (rotor) inlet was determined to be 101334.22 Pa, and the (average) pressure at the (stator) outlet was determined to be 101328.61 Pa. Therefore, $\Delta P = 5.61 \text{ Pa}$.

The second contour of interest pertains to the fluid streamlines as shown below:

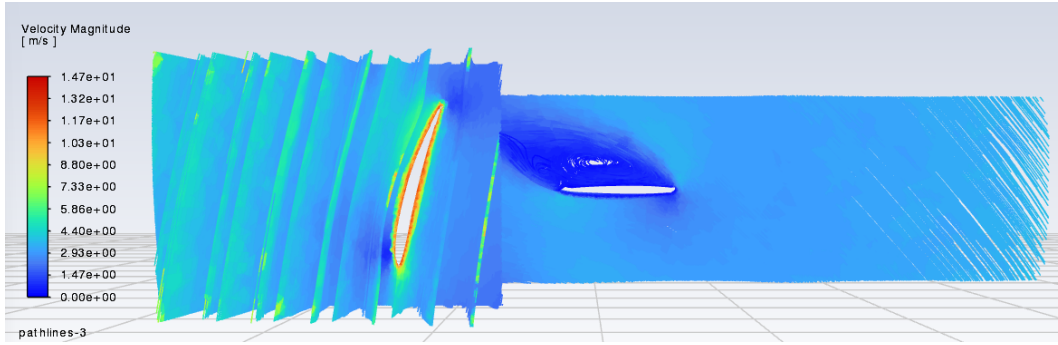


Figure 5: Streamline Contour ($\omega = 1000$ rpm)

The residual, pressure, and streamline contours are shown below that correspond to the setup using the same exact settings as before with the exception of a modified rotor angular speed of...

$\omega = 1500$ rpm

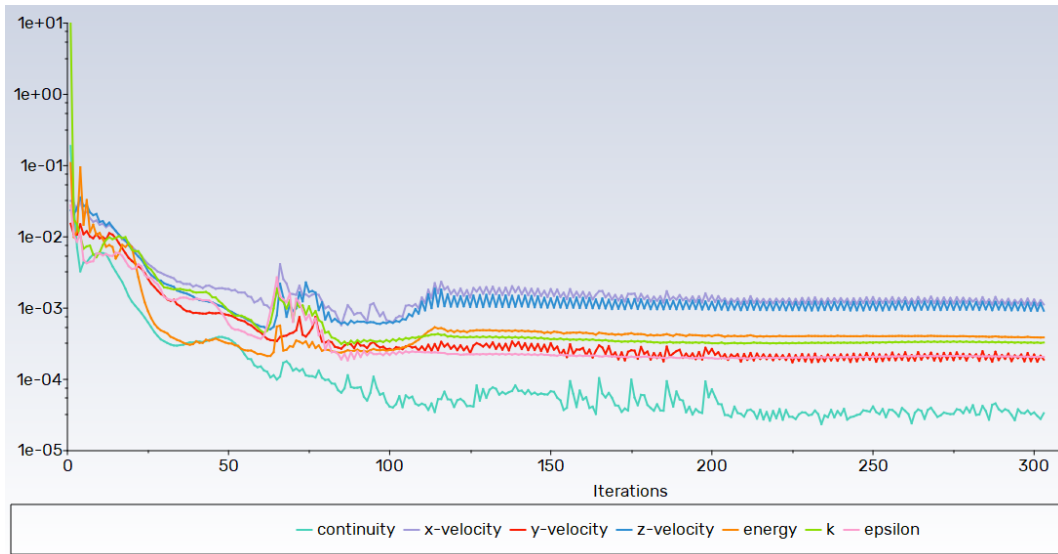


Figure 6: Residual Plot ($\omega = 1500$ rpm)

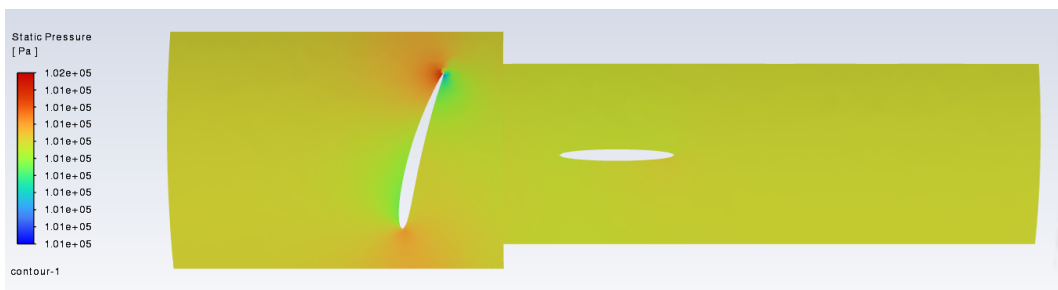


Figure 7: Pressure Contour ($\omega = 1500$ rpm)

The (average) pressure at the (rotor) inlet was determined to be 101336.49 Pa, and the (average) pressure at the (stator) outlet was determined to be 101330.09 Pa. Therefore, $\Delta P = 6.4$ Pa.

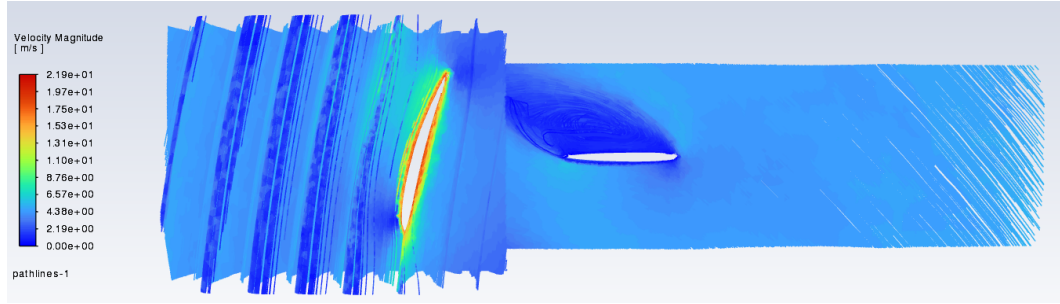


Figure 8: Streamline Contour ($\omega = 1500$ rpm)

The next residual, pressure, and streamline contours are shown below that correspond to the setup using the same exact settings as before with the exception of a modified rotor angular speed of...

$\omega = 2250$ rpm

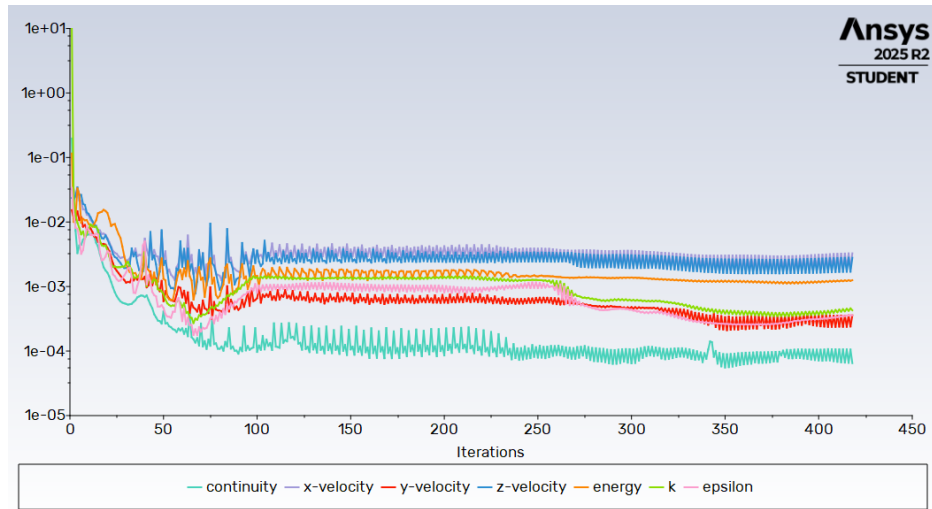


Figure 9: Residual Plots ($\omega = 2250$ rpm)

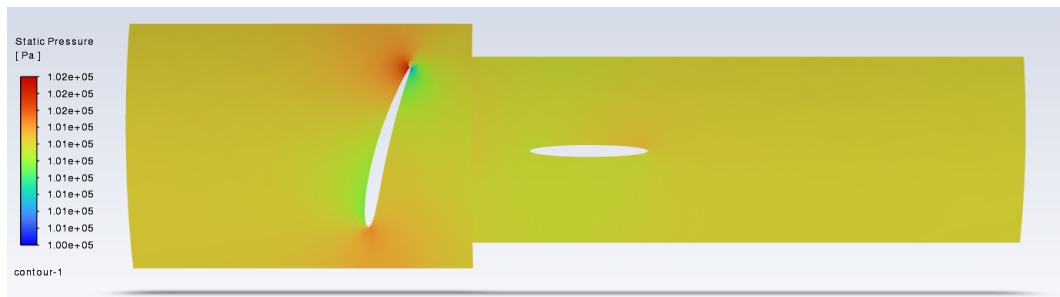


Figure 10: Pressure Contour ($\omega = 2250$ rpm)

The (average) pressure at the (rotor) inlet was determined to be 101341.11 Pa, and the (average) pressure at the (stator) outlet was determined to be 101378.1 Pa. Therefore, $\Delta P = 36.99$ Pa.

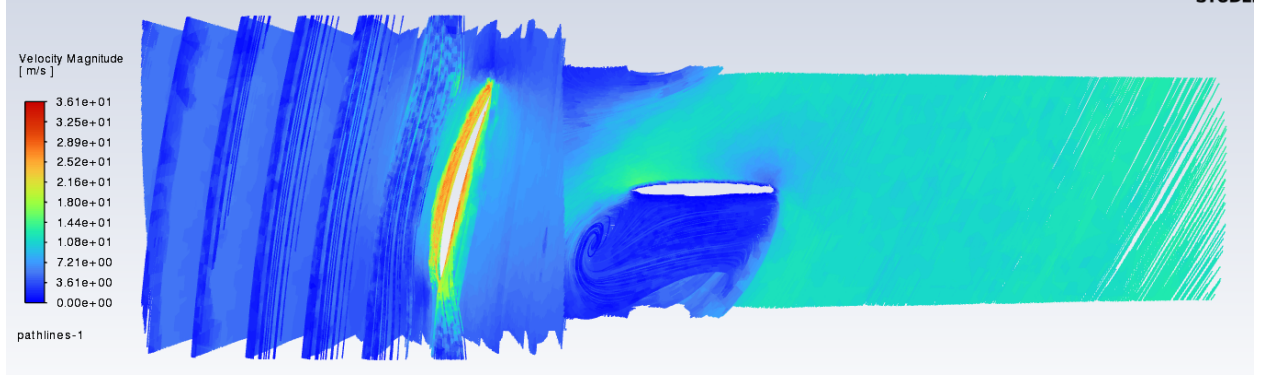


Figure 11: Streamline Contour ($\omega = 2250$ rpm)

Discussion & Tabulation

Overall, the residuals did not get small enough to proclaim the simulation converged adequately. However, the parameter monitors such as inlet/outlet pressure and temperature did converge (trust).

For each angular speed case, the pressure gain across the compressor, the torque applied to the rotor, and the incoming mass flow rate was determined to be given by:

- $(\Delta P)_{1000 \text{ rpm}} = 5.61 \text{ Pa}$ & $\tau_{1000 \text{ rpm}} = 0.00175 \text{ J}$ & $\dot{m}_{1000 \text{ rpm}} = 0.0119 \text{ kg/s}$
- $(\Delta P)_{1500 \text{ rpm}} = 6.4 \text{ Pa}$ & $\tau_{1500 \text{ rpm}} = 0.00401 \text{ J}$ & $\dot{m}_{1500 \text{ rpm}} = 0.0185 \text{ kg/s}$
- $(\Delta P)_{2250 \text{ rpm}} = 36.99 \text{ Pa}$ & $\tau_{2250 \text{ rpm}} = 0.00872 \text{ J}$ & $\dot{m}_{2250 \text{ rpm}} = 0.0301 \text{ kg/s}$

The efficiency of the section as prescribed can be approximated by:

$$\eta = \frac{\dot{m} \Delta P}{\rho_{\text{air}} \tau [\omega]_{\text{rad}}} \quad \text{s.t.} \quad [\omega]_{\text{rad}} = \frac{\pi}{30} [\omega]_{\text{rpm}}$$

Upon plugg'n chugg'n:

- $\eta_{1000 \text{ rpm}} = 29.7\%$
- $\eta_{1500 \text{ rpm}} = 15.3\%$
- $\eta_{2250 \text{ rpm}} = 44.2\%$

It's kind of strange that the efficiency doesn't change monotonically, but that could be due to a miscalculation somewhere, or due to a genuine convergence error. In any case, this is what I've got!